

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 21



By- Vishal Sir

Recap of Previous Lecture



Topic

Algebraic structure



Topic

Semi-group



Topic

Monoid



Topic

Group



Topic

Abelian group / Commutative group



Topic

Practice Questions



Topics to be Covered



Topic

Practice Questions



Topic

Finite group



Topic

Addition modulo 'm' $\oplus_m$



Topic

Multiplication modulo 'm' $\otimes_m$



#Q. Let A be the set of all non-singular matrices of order $n \times n$ over real number and let $*$ be the matrix multiplication operation. Then

determinant $\neq 0$

A

A is closed under $*$ but $\langle A, * \rangle$ is not a semigroup

B

$\langle A, * \rangle$ is a semigroup but not a monoid.

C

$\langle A, * \rangle$ is a monoid but not a group.

D

$\langle A, * \rangle$ is a group but not an abelian group.

✓ ① Closed

$A_{n \times n} \& B_{n \times n}$

then $AB_{n \times n}$

And if $|A| \neq 0 \& |B| \neq 0$

then $|AB| \neq 0$

✓ ② Associative

$$(AB) \cdot C = A \cdot (B \cdot C)$$

#Q. Let A be the set of all non-singular matrices of order $n \times n$ over real number and let $*$ be the matrix multiplication operation. Then

determinant $\neq 0$



A is closed under $*$ but $\langle A, * \rangle$ is not a semigroup



$\langle A, * \rangle$ is a semigroup but not a monoid.



$\langle A, * \rangle$ is a monoid but not a group.



$\langle A, * \rangle$ is a group but not an abelian group.

⑤ Matrix multiplication is not commutative

③ Identity:

$I_{n \times n} \in \text{Set } A$

$B \cdot I = B$

④ Inverse

Inverse will exist for non-singular matrices

And. if A is of order $n \times n$, then $A^{-1}_{n \times n}$

$$|F(S)| = |S|^{|S|}$$

$$f: S \rightarrow S$$

#Q. Let S be any finite set, and $F(S)$ is defined as set of all function on set S . Then $F(S)$ with respect to function composition operation (ie., $\circ$) is.

✓ ① if $f_1: S \rightarrow S$ and $f_2: S \rightarrow S$,
then $f_1 \circ f_2: S \rightarrow S$ and $f_2 \circ f_1: S \rightarrow S$
ie, if $f_1, f_2 \in F(S)$ then $f_1 \circ f_2, f_2 \circ f_1 \in F(S)$

Hence Closed.

A Not a semigroup

B Semi group but not a monoid

✓ ② Function Composition is Associative

✓ **C** Monoid but not a group.

✓ ③ let $f \in F(S)$, $\therefore f: S \rightarrow S$
 $f \circ I_S = f$ and $I_S \circ f = f$

D A group

Here I_S is identity function on set S .

④ Not all functions on set S are bijective. $\therefore$ if f_n is not bijective then Inverse will not exist

$$\text{Max}(\text{smallest of, any other all integers, integer 'a'}) = a$$

#Q. Let Z is the set of all integers. The binary operation $*$ is defined as $a*b = \max(a, b)$ then the structure $(Z, *)$ is

$$a*b = \max(a, b), \forall a, b \in Z$$



Not a semigroup



Semi group but not a monoid



Monoid but not a group.



A group

① for elements $a, b \in Z$, the $\max(a, b)$ will be either a or b
 $\therefore \max(a, b) \in Z$ Hence Closed.

$$\underbrace{\max(\max(a, b), c)}_{\text{Result will be max of three}} = \underbrace{\max(a, \max(b, c))}_{\text{Result will be max of three}}$$

$\therefore$ Associative

$\therefore$ Identity element does not exist

③ Identity element will be smallest integer, but it does not exist

#Q. Let $\mathbb{Q}^+$ be the set of all positive rational numbers. The binary operation $*$ is

defined as $a * b = \frac{ab}{3} \forall a, b \in \mathbb{Q}^+$ If $(\mathbb{Q}^+, *)$ is a group then find

(i) identity element of the group = 3

(ii) inverse of any element $a, \forall a \in \text{Group} = \frac{9}{a}$

(i) let 'e' is identity

$$\therefore a * e = a$$

$$\frac{ae}{3} = a$$

$$\boxed{e = 3}$$

(ii) let $\text{inv}(a)$ is a^{-1}

We know $a * a^{-1} = e$ (identity)

$$\frac{a \cdot a^{-1}}{3} = 3$$

$$\therefore \boxed{a^{-1} = \frac{9}{a}}$$

#Q. Which of the following statement is/are not true.

- A** $\{0, \pm 2k, \pm 4k, \pm 6k, \dots\}$ is a group with respect to addition where k is any fixed positive integer
- B** $\{x \mid x \text{ is real number and } 0 < x \leq 1\}$ is a group with respect to multiplication
- C** $\{2^n \mid n \text{ is an integer}\}$ is a group with respect to multiplication
- D** None of these
- (True)*
- ① Closed ✓*
- ② Associative ✓*
- ③ Identity w.r.t. addition: $0 \in \text{Set}$*
- ④ $\text{inv}(0)=0, \text{inv}(2k)=-2k, \text{inv}(4k)=-4k, \dots$*

#Q. Which of the following statement is/are not true.

A $\{0, \pm 2k, \pm 4k, \pm 6k, \dots\}$ is a group with respect to addition where k is any fixed positive integer

B $\{x \mid x \text{ is real number and } 0 < x \leq 1\}$ is a group with respect to multiplication

C $\{2^n \mid n \text{ is an integer}\}$ is a group with respect to multiplication

D None of these

False

① Closed

② Multiplication is associative

③ Identity w.r.t. multiplication = 1 and it is a member of the set

④ $\text{inv}(1) = 1$, and inverse does not exist for any other element of set

$\therefore$ given set is a monoid but not a group w.r.t. multiplication

#Q. Which of the following statement is/are not true.

$$\{ \dots, 2^{-2}, 2^{-1}, 2^0, 2^1, 2^2, 2^3, \dots \}$$

① Closed

let n_1, n_2 are two integers
then $2^{n_1}, 2^{n_2} \in \text{Set}$

$$2^{n_1} \cdot 2^{n_2} = 2^{n_1+n_2}$$

if n_1, n_2 are integers then n_1+n_2 is also an integer
 $\therefore 2^{n_1+n_2} \in \text{Set}$

A

$\{0, \pm 2k, \pm 4k, \pm 6k, \dots\}$ is a group with respect to addition where k is any fixed positive integer

B

$\{x \mid x \text{ is real number and } 0 < x \leq 1\}$ is a group with respect to multiplication

C

$\{2^n \mid n \text{ is an integer}\}$ is a group with respect to multiplication

② Multiplication is associative

③ Identity = 1 = $2^0 \in \text{Set}$

④ Inverse

let $2^{n_1} \in \text{Set}$, then $\text{inv}(2^{n_1}) = ?$
 $2^{n_1} \cdot \text{inv}(2^{n_1}) = 1 \Rightarrow \text{inv}(2^{n_1}) = \frac{1}{2^{n_1}}$
 $\text{inv}(2^{n_1}) = 2^{-n_1} \in \text{Set}$

D

None of these

True



Topic : Finite Group

- * The group $(G, *)$ is called a finite group if " G " is a finite set.
- * Let $(G, *)$ is a finite group, then order of group $(G, *)$ is defined as the number of elements in set G .
- * For finite group $(G, *)$, Order of group G can be denoted by $|G|$ or $O(G)$



Topic : Finite Group

- ① The set $\{0\}$ is a finite group of order = 1 with respect to binary operation Addition
- ② The set $\{1\}$ is a finite group of order = 1 with respect to binary operation Multiplication
- ③ The set $\{-1, 1\}$ is a finite group of order = 2 with respect to binary operation Multiplication

These are the only three finite groups of real numbers with respect to binary opⁿ addition and/or multiplication



Topic : NOTE

- ① If ' e ' is the identity element w.r.t. binary operation ' $*$ ', then $\{e\}$ is always a finite group of order = 1 w.r.t. binary operation ' $*$ '.
- ② In a finite group of order = 2.
Every element of the group is inverse of itself.



Topic : NOTE

③ Let 'S' is any non-Empty Set.

Set 'S' with respect to binary operation $*$ is a group if and only if ① $a * b^{-1} \in S, \forall a, b \in S$ and ② ' $*$ ' is associative

ie: Statement ① should be enough to prove that

- (i) Set S is closed w.r.t. opⁿ $*$
- (ii) identity w.r.t. opⁿ $*$ is present in Set S.
- (iii) Inverse of every element of Set S is present in Set S

* Let $a * b^{-1} \in S, \forall a, b \in S$



① Identity
Prove that identity $\in S$

Let $a \in S$
i.e. $a, a \in S$
We know.
if $a, a \in S$
then $a * a^{-1} \in S$
(identity) e
 $\therefore$ (identity) $e \in S$

② Inverse
if $a \in S$, then
Prove that $a^{-1} \in S$

* We have already proved that identity $e \in S$
* if 'e' is the only element of the set then $\{e\}$ w.r.t '*' is guaranteed to be a group
* if there exist any other element 'a' in set S then $e, a \in S$
We know
if $e, a \in S$
then $e * a^{-1} \in S$
 $\downarrow$
 $a^{-1} \in S$

③ Closed
if $a, b \in S$, then Prove that
 $a * b \in S$

* Let $a, b \in S$
* if $b \in S$ then $b^{-1} \in S$
 $\therefore$ if $a, b \in S$, then $a, b^{-1} \in S$
* if $a, b^{-1} \in S$
then $a * (b^{-1})^{-1} \in S$
 $\downarrow$
 $a * b \in S$
i.e. if $a, b \in S$
then $a * b \in S$



Topic : Finite Group

We know,

Cube roots of unity are $1, \omega$ and ω^2



and

$$\omega^3 = 1$$

The set $\{1, \omega, \omega^2\}$ form a finite group of order = 3 with respect to binary opⁿ multiplication

- ✓ ① Closed
- ✓ ② Associative
- ✓ ③ Identity
- ④ Inverse ?

↳ in order to identify the inverse we can use Composition table

Binary opⁿ (multiplication)

	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	$\omega^3 = 1 = e$
ω^2	ω^2	$\omega^3 = 1 = e$	$\omega^4 = \omega$

Annotations from the table:

- $\text{inv}(1) = 1$ (pointing to the top-left cell)
- $\text{inv}(\omega) = \omega^2$ (pointing to the cell containing ω^2 in the ω row)
- $\text{inv}(\omega^2) = \omega$ (pointing to the cell containing ω in the ω^2 row)

inverse also exist for every element of set in group

eg:- Four roots of unity are $1, -1, i$ and $-i$



We know $i^2 = -1 \Rightarrow i = \sqrt{-1}$

• The set $\{1, -1, i, -i\}$ form a finite group of order=4 with respect to multiplication

✓ ① Closed

✓ ② Associative

✓ ③ Identity

④ Inverse:

$$\text{inv}(1) = 1$$

$$\text{inv}(-1) = -1$$

$$\text{inv}(i) = -i$$

$$\text{inv}(-i) = i$$

$\cdot$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	$i^2 = -1$	$-i^2 = -(-1) = 1$
-i	-i	i	$-i^2 = 1$	$i^2 = -1$

The set of ' n ' roots of unity will form a finite group of order= n with respect to binary operation "multiplication"



Topic : Addition modulo ' m ' $\oplus_m$

it gives remainder

- Let ' m ' is any fixed positive integer
for any two non-negative integers a and b
we define,

$$a \oplus_m b = \begin{cases} (a+b) & , \text{ if } (a+b) < m \\ \pi & , \text{ otherwise \{ i.e., } (a+b) \geq m \} \end{cases}$$

$$\begin{matrix} \text{|||} \\ (a+b) \% m \end{matrix}$$

it gives
remainder

Where ' π ' is the remainder obtained
On dividing $(a+b)$ by ' m '.

eg. $2 \oplus_7 4 = 6$
 $3 \oplus_7 5 = 1$



Topic : Multiplication modulo 'm' $\otimes_m$

Let 'm' is any fixed positive integer
for any two non-negative integers a and b
we define.

$$a \otimes_m b = \begin{cases} ab & \text{if } ab < m \end{cases}$$

$$\text{eg. } 2 \otimes_8 3 = 6 \quad \text{|||} \quad (a.b) \% m$$

$$3 \otimes_8 5 = 7$$

Otherwise (ie $ab \geq m$)
where 'r' is the remainder obtained
on dividing (a.b) by 'm'.



2 mins Summary



Topic

Practice Questions

Topic

Finite group

Topic

Addition modulo 'm' $\oplus_m$

Topic

Multiplication modulo 'm' $\otimes_m$

THANK - YOU